

SiteSupervise & Tersus MVP S1 Integration

Weekly Status Report

Date:	August 3, 2026	Period	Week 31 (Jul 28 - Aug 3, 2026)
Project	SiteSupervise - Tersus MVP S1 Handheld Scanner Integration		

Executive Summary

This week, the foundational backend architecture for the **Tersus MVP S1 Handheld Scanner Integration** into **SiteSupervise** was successfully established. Key accomplishments include setting up domain-driven Django application modules, implementing core scan session ingestion endpoints with direct-to-S3 pre-signed URL workflow, configuring OpenAPI 3.0 auto-generation via drf-spectacular, and establishing robust unit tests covering end-to-end scan ingestion.

1. Accomplishments & Key Deliverables

A. Backend Architecture & Domain Setup

- Scaffolded Clean Architecture directory structure separating domain logic into clean modules: scans, reports, projects, inspections, storage, processing, notifications, common, authentication, and audit.
- Configured PostgreSQL database integration and Django migrations for ScanSession and ScanMetadata models.
- Set up containerization with docker-compose.yml defining web, db (PostgreSQL), and redis services.

B. Core Scan Ingestion API Pipeline

- **Session Initialization Endpoint** (POST /api/v1/scans/session/): Accepts scanner telemetry and expected upload size, returning unique UUID session_id and pre-signed upload target URL (upload_url).
- **Metadata Ingestion Endpoint** (POST /api/v1/scans/{session_id}/metadata/): Accepts spatial telemetry (latitude, longitude, elevation), operator ID, and field notes.
- **Upload Finalization Endpoint** (POST /api/v1/scans/{session_id}/finalize/): Validates current status and transitions scan session to 'processing', queuing job for background Celery processing pipelines.

C. OpenAPI Specification & Schema Automation

- Integrated drf-spectacular for auto-generating OpenAPI 3.0 definitions.
- Resolved schema warnings and type annotations across serializer fields and API views.
- Exported complete api.json OpenAPI schema file for client integration and frontend code generation.

D. Verification & Testing

- Developed suite of unit tests in apps/scans/tests.py using Django REST Framework testing suite.
- Validated all 6 test scenarios (Session creation, Metadata attachment, Finalization status flow, Invalid transitions, duplicate metadata rejections). All tests passed with 100% success rate.

2. API Documentation Overview

Endpoint	Method	Auth Required	Description
/api/v1/auth/token/	POST	No	Generate JWT access and refresh token pair
/api/v1/auth/token/refresh/	POST	No	Refresh expired JWT access token
/api/v1/scans/session/	POST	Yes (JWT)	Initialize scan upload session & obtain S3 URL
/api/v1/scans/{session_id}/metadata/	POST	Yes (JWT)	Attach GPS/sensor metadata & operator notes
/api/v1/scans/{session_id}/finalize/	POST	Yes (JWT)	Finalize upload and trigger background processing
/api/schema/	GET	No	Fetch raw OpenAPI 3.0 schema
/api/docs/	GET	No	Interactive Swagger UI API documentation

3. How to Generate API Documentation & Artifacts

1. Generating api.json (OpenAPI 3.0 Schema)

Run the following command from the backend/ directory:

```
python manage.py spectacular --file api.json
```

2. Interactive Swagger UI

Start the local server or Docker container and navigate to:

- **Swagger UI:** <http://localhost:8000/api/docs/>
- **Schema JSON:** <http://localhost:8000/api/schema/>

3. Running Automated Tests

```
python manage.py test
```

4. Next Week's Objectives & Roadmap

- 1. Celery & Background Processing:** Connect FinalizeUploadView trigger to asynchronous Celery tasks for point cloud and thermal image processing.
- 2. S3 Direct Upload Integration:** Replace mock S3 URL generation with actual AWS S3 / MinIO pre-signed upload URL signatures (boto3).

- 3. Data Fusion & Spatial Alignment:** Implement service layer logic for aligning Tersus LiDAR point clouds with SiteSupervise BIM coordinates (data_fusion_design.md).
- 4. Automated QA/QC PDF Reporting:** Scaffold report generation templates in apps/reports for weekly automated QA/QC summaries.